

CSE340: Computer Architecture

Quiz 3

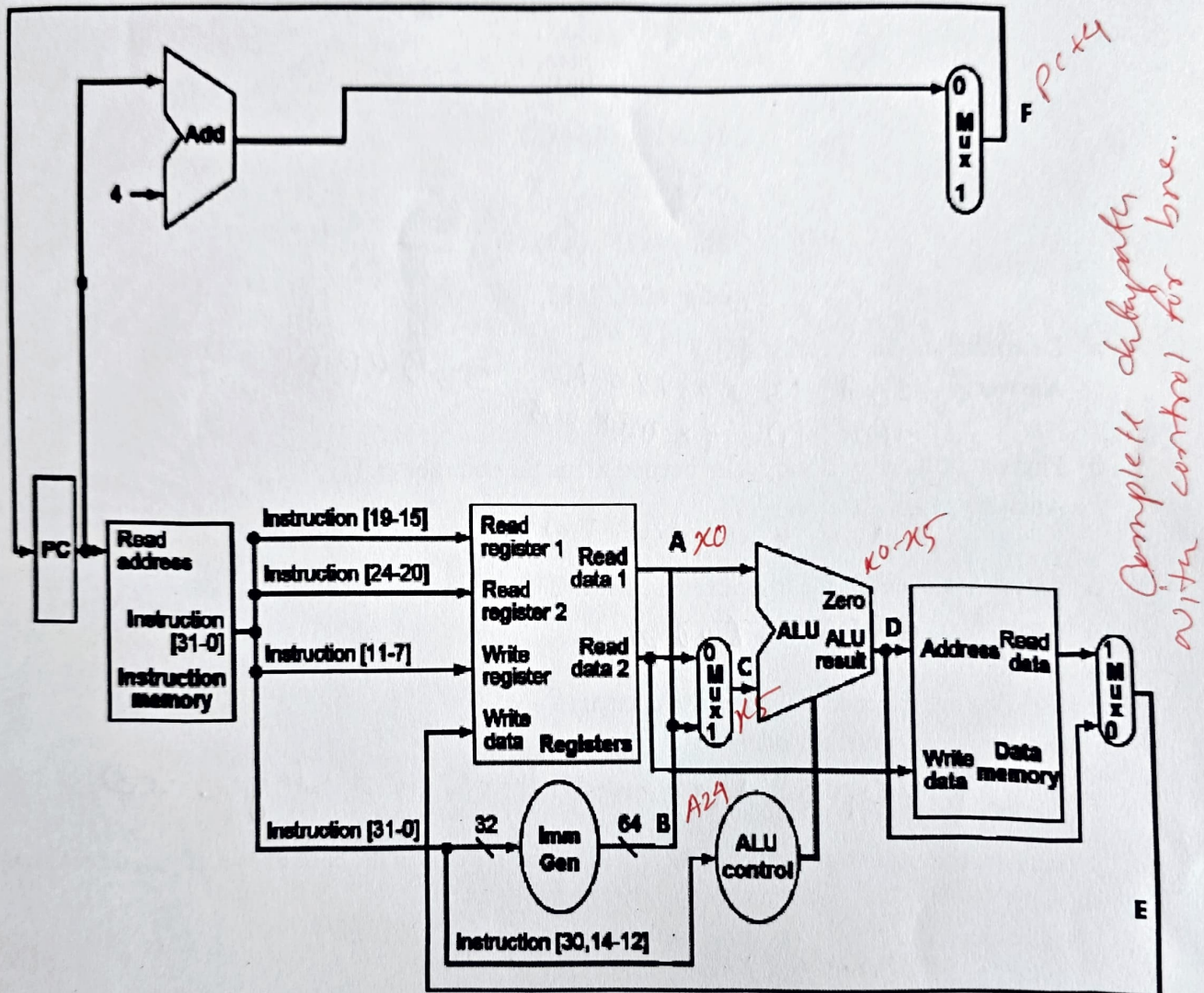
Time: 40 minutes

Semester: Summer 26

Total Marks: 20

Name: _____ ID: _____ Section: _____

Question 1 [10]



Consider the single cycle datapath above.

- Complete the missing parts of the datapath in the figure above so the BNE works. [4]
- Find the following values for the given instruction. Assume, PC = 0xA25, X5 = 0x50D. [6]

Instruction	A	B	C	D	E	F
BNE X0, X5, 0xA24	0x0	0xA29	0x50D	0x50D	X	0xA29
(Your answers must be in hexadecimal)	Branch	MemRead	MemWrite	MemToReg	ALUSrc	RegWrite
	1	0	0	X	0	0

Question 2 [6]

In the same datapath from question 1, consider the following information.

Process	Instruction Fetch	Register read/write	ALU operation	Memory access
Time	20ps	15ps	20ps	20ps

and x18, x19, x7

srl x19, x18, x5

ld x19, 16(x5)

or x5, x6, x7

addi x19, x5, 5

add x19, x18, x7

- a. Determine the clock period. [1]

Answer:

$$20 + 15 + 20 + 20 + 15 = 90 \text{ ps}$$

IF ID EX MEM WB

- b. Find out how many clock cycles needed to run the code above. [1]

Answer:

6 for 6 instructions.

- c. Calculate the CPU time to execute the code. [1]

Answer:

$$6 \times 90 = 540 \text{ ps}$$

If we are executing the code in a pipelined datapath,

- d. Determine the clock period. [1]

Answer:

20ps for max time in a stage

- e. Analyze the code to determine if there will be any hazards in the code above. If yes, detect how many. [1+2]

Answer:

Yes. 2.

- f. If the hazards are solved without using any stalls, how many clock cycles are needed to execute the code. [1]

Answer:

10

- g. Calculate the CPU time to execute the code. [1]

Answer:

200ps

- h. Find out how much improvement to the execution time we got because of pipelining. [1]

Answer:

$$340 \text{ ps} = (540 - 200) \text{ ps}$$

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Question 1 [10]

The diagram illustrates a processor datapath with the following components and connections:

- PC (Program Counter):** Outputs to the **Read address** of **Instruction memory** and to an **Add** block.
- Instruction memory:** Receives the **Read address** and outputs instruction fields: **Instruction [19-15]**, **Instruction [24-20]**, **Instruction [11-7]**, and the full **Instruction [31-0]**.
- Registers:** A block containing **Read register 1**, **Read register 2**, and **Write register**. It receives **Instruction [19-15]** and **Instruction [24-20]** for reading, and **Instruction [11-7]** for writing. It outputs **Read data 1** and **Read data 2**.
- Imm Gen (Immediate Generator):** Takes **Instruction [31-0]** and **Instruction [30,14-12]** as inputs. It outputs a 32-bit value **B** and a 64-bit value **B** (labeled **x16**).
- ALU control:** Receives the 64-bit **B** and the **Instruction [30,14-12]** field. It outputs control signals to the **ALU**.
- ALU (Arithmetic Logic Unit):** Receives **Read data 1** (labeled **A x6**), **Read data 2** (labeled **C**), and control signals. It outputs the **ALU result** (labeled **D**) and a **Zero** flag.
- Mux 1 (Multiplexer 1):** Selects between **Read data 1** and **Read data 2** based on the **Write register** field. Its output is **Write data**.
- Mux 2 (Multiplexer 2):** Selects between the **ALU result** and the output of **Mux 1** based on the **Write register** field. Its output is **Write data**.
- Shift left 1:** Takes the **ALU result** and shifts it left by one bit. Its output is **Sum**.
- Add Sum:** Adds the **Sum** to the **PC** output. Its output is **PC+4**.
- Mux 3 (Multiplexer 3):** Selects between the **ALU result** and the output of **Add Sum** based on the **Zero** flag. Its output is **F**.
- Mux 4 (Multiplexer 4):** Selects between the **ALU result** and the output of **Add Sum** based on the **Zero** flag. Its output is **E**.

Handwritten red notes and labels in the diagram include:

- PC+4** (near the top right)
- complete datapath with control** (written vertically on the right)
- A x6** (near the ALU input)
- B x16** (near the Imm Gen output)
- x6 + 0x10** (near the ALU output)
- Zero** (near the ALU output)

- Complete the missing parts of the datapath in the figure above. [4]
- Find the following values for the given instruction. Assume, PC = 0xB25, X6 = 0x60D. [6]

Instruction	A	B	C	D	E (Don't know)	F
SD X6, 16(X6)	0x60D	0x10	0x10	0x61D	X	0xB20
(Your answers must be in hexadecimal)	Branch	MemRead	MemWrite	MemToReg	ALUSrc	RegWrite
	0	0	1	X	1	0

Question 2 [6]

In the same datapath from question 1, consider the following information.

Process	Instruction Fetch	Register read/write	ALU operation	Memory access
Time	35ps	15ps	20ps	25ps

mul x20, x21, x7

andi x21, x20, 5

add x21, x5, x6

or x5, x6, x7

ld x21, 16(x5)

addi x21, x20, 7

- a. Determine the clock period. [1]

Answer:

$$35 + 15 + 20 + 25 + 15 = 110\text{ps}$$

- b. Find out how many clock cycles needed to run the code above. [1]

Answer:

6 for 6 instructions.

- c. Calculate the CPU time to execute the code. [1]

Answer:

$$6 \times 110 = 660\text{ps}$$

If we are executing the code in a pipelined datapath,

- d. Determine the clock period. [1]

Answer:

35ps for max time in one stage.

- e. Analyze the code to determine if there will be any hazards in the code above. If yes, detect how many. [1+2]

Answer:

Yes. 2.

- f. If the hazards are solved without using any stalls, how many clock cycles are needed to execute the code. [1]

Answer:

10.

- g. Calculate the CPU time to execute the code. [1]

Answer:

$$35 \times 10 = 350\text{ps}$$

- h. Find out how much improvement to the execution time we got because of pipelining. [1]

Answer:

$$660 - 350 = 310\text{ps}$$

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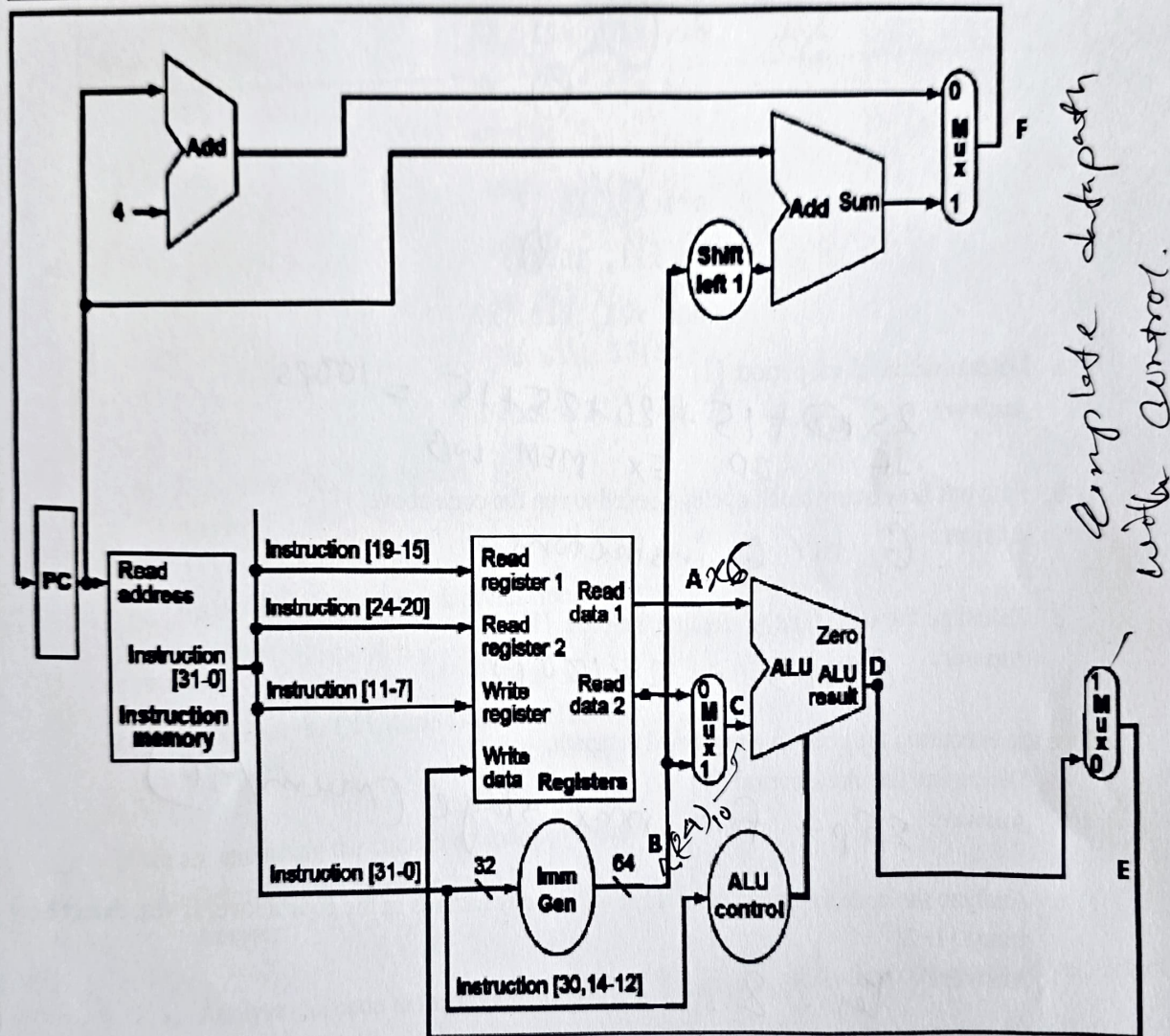
Total Marks: 20

Name: _____

ID:

Section:

Question 1 [10]



Consider the single cycle datapath above.

- Complete the missing parts of the datapath in the figure above. [4]
- Find the following values for the given instruction. Assume, PC = 0xB24, X6 = 0x60C. [6]

Instruction	A	B	C	D	E (Don't know)	F
LD X5, 24(X6)	0x60C	0x18	0x18	0x624	X	0x28
(Your answers must be in hexadecimal)	Branch	MemRead	MemWrite	MemToReg	ALUSrc	RegWrite
	0	1	0	1	1	1

Question 2 [6]

In the same datapath from question 1, consider the following information.

Process	Instruction Fetch	Register read/write	ALU operation	Memory access
Time	25ps	15ps	20ps	25ps

div(x20, x21, x7
and x21, x20, x5
addi x21, x5, 16
ori x5, x6, 7
ld x21, 16(x5)
add x21, x20, x7

- a. Determine the clock period. [1]

Answer: $25 + 15 + 20 + 25 + 15 = 100\text{ps}$
IF ID EX MEM WB

- b. Find out how many clock cycles needed to run the code above. [1]

Answer: 6 for 6 instructions

- c. Calculate the CPU time to execute the code. [1]

Answer: $600\text{ps} = 6 \times 100\text{ps}$

If we are executing the code in a pipelined datapath,

- d. Determine the clock period. [1]

Answer: 25ps for max stage (mem/IF)

- e. Analyze the code to determine if there will be any hazards in the code above. If yes, detect how many. [1+2]

Answer: Yes. 2.

- f. If the hazards are solved without using any stalls, how many clock cycles are needed to execute the code. [1]

Answer: 10

- g. Calculate the CPU time to execute the code. [1]

Answer: $25 \times 10 = 250\text{ps}$

- h. Find out how much improvement to the execution time we got because of pipelining. [1]

Answer: $600 - 250\text{ps} = 350\text{ps}$

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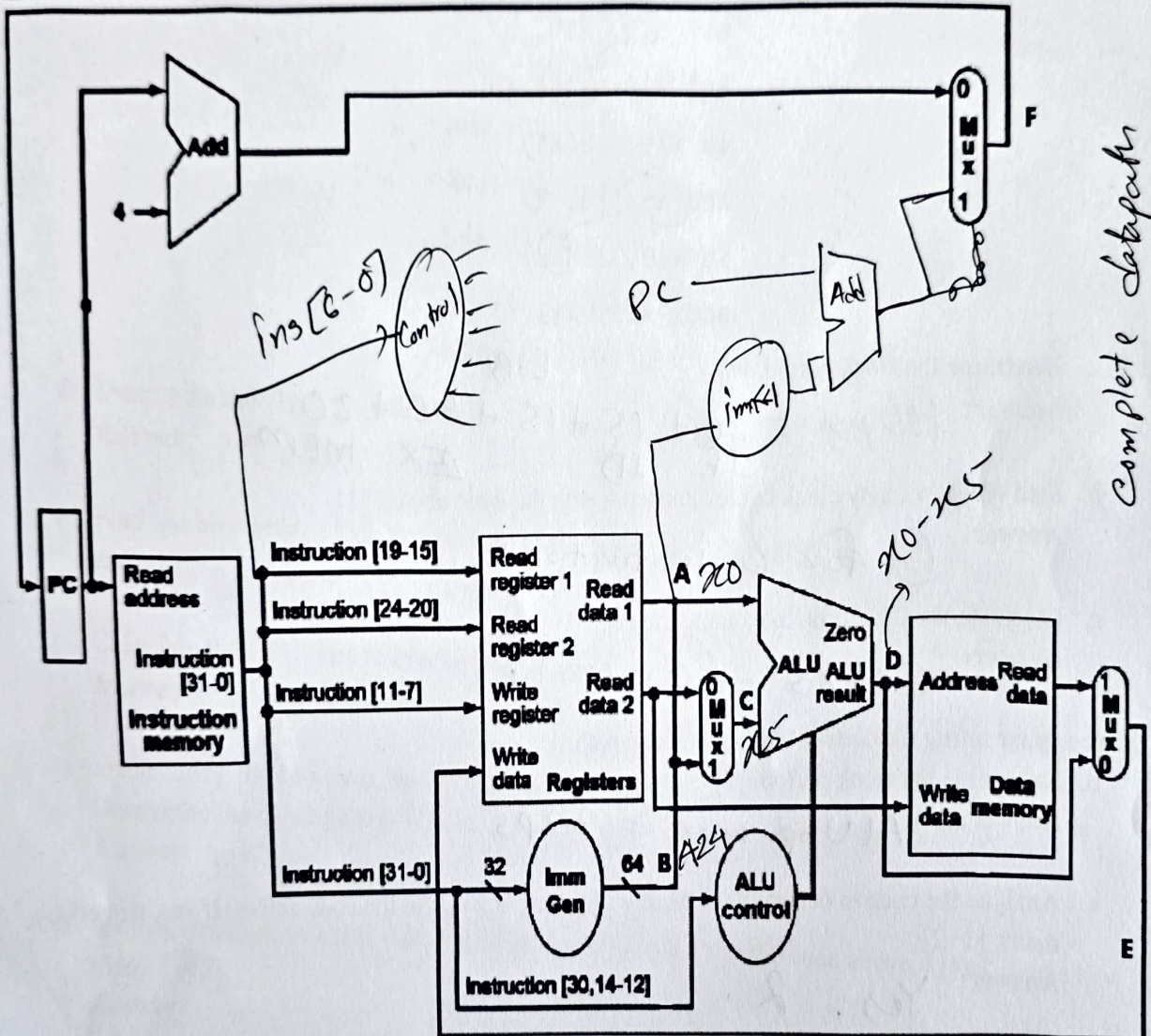
Total Marks: 20

Name: Solve

ID: _____

Section: 29

Question 1 [10]



Complete datapath with control.

Consider the single cycle datapath above.

- Complete the missing parts of the datapath in the figure above. [4]
- Find the following values for the given instruction. Assume, PC = 0xA24, X5 = 0x50C. [6]

Instruction	A	B	C	D	E	F
BEQ X0, X5, 0xA24	0	0xA24	0x50C	0x50C	X	0xA28
(Your answers must be in hexadecimal)	Branch	MemRead	MemWrite	MemToReg	ALUSrc	RegWrite
	1	0	0	X	0	0

Question 2 [6]

In the same datapath from question 1, consider the following information.

Process	Instruction Fetch	Register read/write	ALU operation	Memory access
Time	20ps	15ps	30ps	20ps

```
mul x18, x19, x7
srl x19, x18, x5
ld x19, 16(x5)
and x5, x6, x7
ld x19, 16(x5)
addi x19, x18, 7
```

- a. Determine the clock period. [1]

Answer: $100 \text{ ps} = 20 + 15 + 15 + 30 + 20$
IF ID EX MEM WB

- b. Find out how many clock cycles needed to run the code above. [1]

Answer: 6 for 6 instructions.

- c. Calculate the CPU time to execute the code. [1]

Answer: $600 \text{ ps} = (6 \times 100) \text{ ps}$

If we are executing the code in a pipelined datapath,

- d. Determine the clock period. [1]

Answer: ALU op max = 30 ps

- e. Analyze the code to determine if there will be any hazards in the code above. If yes, detect how many. [1+2]

Answer: Yes, 2.

- f. If the hazards are solved without using any stalls, how many clock cycles are needed to execute the code. [1]

Answer: 10

- g. Calculate the CPU time to execute the code. [1]

Answer: $10 \times 30 = 300 \text{ ps}$

- h. Find out how much improvement to the execution time we got because of pipelining. [1]

Answer: $600 - 300 = 300 \text{ ps}$ improvement.